

Exercise 4

```
|: import os
import openai
import tiktoken
from dotenv import load_dotenv, find_dotenv
_ = load_dotenv(find_dotenv()) # read local .env file

openai.api_key = os.environ['OPENAI_API_KEY']
```

```
In [17]: # Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You are a helpful assistant with deep knowledge in physics."}

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "Explain how black holes are formed."}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

Black holes are formed when a massive star runs out of fuel and collapses in on itself during a supernova explosion. As the star collapses, the outer layers are blasted into space while the core collapses under its own gravity. If the core has a mass greater than about three times that of the Sun, it will collapse into a black hole.

The core of the star becomes so dense that its gravity becomes incredibly strong, causing it to collapse to a point of infinite density called a singularity. This singularity is surrounded by an event horizon, which is the point of no return for anything that crosses it. Once an object or light crosses the event horizon, it can never escape the gravitational pull of the black hole.

Black holes come in different sizes, with stellar-mass black holes being formed from the collapse of massive stars, and supermassive black holes found at the centers of galaxies, with masses millions or even billions of times that of the Sun.

```
# Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You are a helpful assistant with deep knowledge in Sri Lanka"}

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "I want to plan a 4-day family trip to Kandy, Sri Lanka"}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

Based on your budget, I will suggest some affordable options for your trip to Kandy, Sri Lanka.

****Accommodation:****

- Hotel Topaz: A budget-friendly hotel located in Kandy offering comfortable rooms and amenities for around Rs. 5,000 per night.

****Places to Visit:****

Day 1:

- Visit the Temple of the Tooth Relic
- Explore the Royal Botanical Gardens
- Walk around Kandy Lake

Day 2:

- Visit the Udawattakele Forest Reserve
- Explore the Bahirawakanda Vihara Buddha Statue
- Visit the Kandy View Point

Day 3:

- Explore the Peradeniya Railway Station
- Visit the Commonwealth War Cemetery
- Experience a traditional Kandyan dance show

Day 4:

- Visit the Ceylon Tea Museum
- Explore the Kandy Garrison Cemetery
- Go shopping at Kandy City Center

****Transportation:****

- Rent a car for convenience and flexibility, costing around Rs. 4,000 per day.
- Alternatively, you can use tuk-tuks or public transport for shorter distances.

****Local Food:****

- Try traditional Sri Lankan dishes like rice and curry, kottu roti, hoppers, and pol sambol at local eateries and street food stalls.

****Estimated Expenses:****

- Accommodation: Rs. 5,000 x 3 nights = Rs. 15,000
- Food: Rs. 2,000 per day x 4 days = Rs. 8,000
- Transportation: Rs. 4,000 per day x 4 days = Rs. 16,000
- Entrance fees and activities: Rs. 2,000 per person for the entire trip = Rs. 8,000
- Total estimated expenses: Rs. 47,000

This estimated budget can be adjusted based on your preferences and any additional activities or souvenirs you plan to purchase during your trip. Enjoy your family vacation to Kandy!

```
# Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You are a helpful assistant with deep knowledge in"}

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "I want to plan a 3-day beach vacation to Mirissa, Sri"}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

That sounds like a fantastic beach vacation plan! Mirissa in Sri Lanka is a beautiful destination known for its stunning beaches and water activities. Here's a detailed itinerary for your 3-day trip:

Day 1:

- **Morning**: Arrival in Mirissa

- Check into affordable accommodation options like hostels or guesthouses. Some recommendations are Hangover Hostels Mirissa or Paris Guest House.

- ****Afternoon****:

- Head to Mirissa Beach for some relaxing time by the sea.
 - Enjoy water activities like swimming and snorkeling.

- ****Evening****:

- Have dinner at a local restaurant like Zephyr Restaurant with seafood special ties.

- Watch the sunset at Coconut Tree Hill for a breathtaking view.

Day 2:

- ****Morning****:

- Have breakfast at a local café like Dewmini Roti Shop.
 - Visit Secret Beach for a more secluded beach experience.

- ****Afternoon****:

- Try out surfing lessons at Weligama Beach.
 - Relax and soak up the sun.

- ****Evening****:

- Enjoy a beachside BBQ dinner at Sun n' Sea Restaurant.

- Explore the nightlife at popular spots like Ceysands or Red Lobster Night Club.

Day 3:

- ****Morning****:

- Take a whale watching tour from Mirissa Harbour.

- ****Afternoon****:

- Explore nearby attractions like Parrot Rock Bridge and Matara Fort.

- ****Evening****:

- Have a farewell dinner at Shady Lane Restaurant.

- Head back to your accommodation to pack and prepare for departure.

Estimated Costs:

- Accommodation: \$15-30 per night per person

- Transportation: Tuk-tuks are affordable for short distances, consider hiring a van for day trips

- Activities: \$20-50 per activity per person

- Food: \$5-15 per meal per person in local restaurants

Tips:

- Book accommodation in advance for better deals.
- Bargain with tuk-tuk drivers for transportation costs.
- Carry sunscreen, hats, and water bottles for outdoor activities.

I hope this itinerary helps you plan a wonderful beach vacation in Mirissa, Sri Lanka! Let me know if you need any more details or assistance.